

Cybersecurity Certification Roadmap Chart Component

Overview

Your task is to implement an interactive chart component that visualizes cybersecurity certifications based on community ratings. This component is inspired by the [certification roadmap](#) found on CyberDefenders.



The chart displays certifications on a scatter plot where:

- **X-axis (Market Presence):** Represents the total number of votes/ratings a certification has received
- **Y-axis (Satisfaction):** Represents the average satisfaction score across all votes

Technical Requirements

- **Framework:** Next.js
- **UI Library:** shadcn/ui (dark theme)
- **Language:** TypeScript
- **Source Code:** Complete Next.js project with:
 - Chart component implementation
 - Filter and search components
 - Responsive layout
 - Type definitions
- **README.md** including:
 - Setup instructions
 - Architecture decisions and rationale
 - Any assumptions made
 - Known limitations or trade-offs
- **Git Repository:** With a meaningful commit history showing your development process

Bouns:

- Unit tests for key components
 - Animation/transitions on chart interactions
 - Accessibility considerations (ARIA labels, keyboard navigation)
 - Performance optimizations for rendering many data points
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Functional Requirements

1. Chart Visualization

Implement an interactive scatter plot chart that:

- Plots each certification as a point/bubble on the chart
- X-axis: Market presence / popularity (based on `market_presence` field)
- Y-axis: Community satisfaction score (based on `satisfaction` field)

2. Color Coding by Certification Type

Certifications should be color-coded based on their `cert_type`:

- `blue` - Blue Team certifications (defensive security)
- `red` - Red Team certifications (offensive security)
- `infoSec` - General Information Security certifications

3. Interactive Features

- **Hover state:** Display a tooltip showing certification details (name, total_votes)
- **Click interaction:** Open a detail panel or modal with full certification information
- **Filtering:** Allow users to filter certifications by:
 - Certification type (blue/red/infoSec)
 - Skill level (Novice, Beginner, Intermediate, Advanced, Expert)
- **Search:** Implement a search functionality to find certifications by name or abbreviation
- **Full screen:** Add support for showing the chart in full-screen mode.

4. Responsive Design

- The chart must be fully responsive
 - Consider mobile-specific interactions (tap instead of hover)
 - Ensure readability on smaller screens
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Data Structure

Use the following API endpoint structure for your implementation. **You can mock this data locally.**

An **array** of certification objects with the following structure:

```
{
  id: number;
  slug: string;
  title: string;
  abbreviation: string;
  description: string;
  image: string;
  url: string;
  cost: string;
  training_included: boolean;
  number_of_attempts: number;
  job_roles_titles: string[];
  cert_type: "blue" | "red" | "infoSec";
  total_votes: number;
  market_presence: number; // 0.0 to 1.0
  cost_effectiveness: number; // 1.0 to 5.0
  skill_level: "Novice" | "Beginner" | "Intermediate" | "Advanced" | "Expert";
  quality: number; // 1.0 to 5.0
  satisfaction: number; // 1.0 to 5.0
  provider: {
    name: string;
    url: string;
    image: string;
  } | null;
  domains_covered_titles: string[];
  requirements_data: {
    knowledge: string;
    work_experience: string;
    prior_courses_and_certifications: string;
  };
  exam_details_data: {
    format: string;
    duration: string | null;
    report_required: boolean;
  };
  valid_for: string | null;
}
```

Submission

1. Link to your Git repository (GitHub)
2. Link to a deployed version (Vercel, Netlify, or similar) - optional but appreciated
3. Any additional notes or documentation you'd like to share

Deadline: Wednesday, Jan 21 (23:59)

Good luck! We look forward to seeing your implementation.